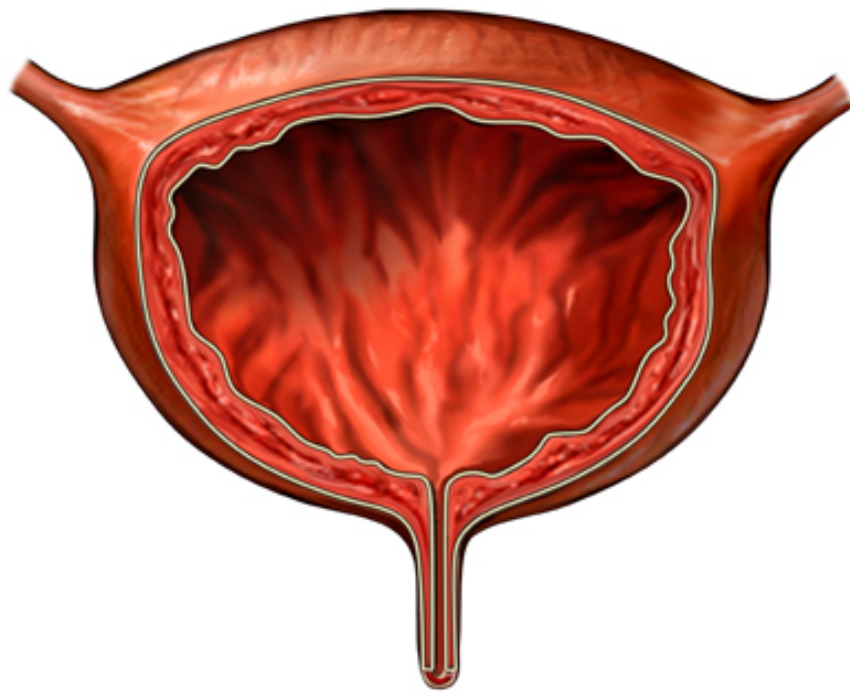


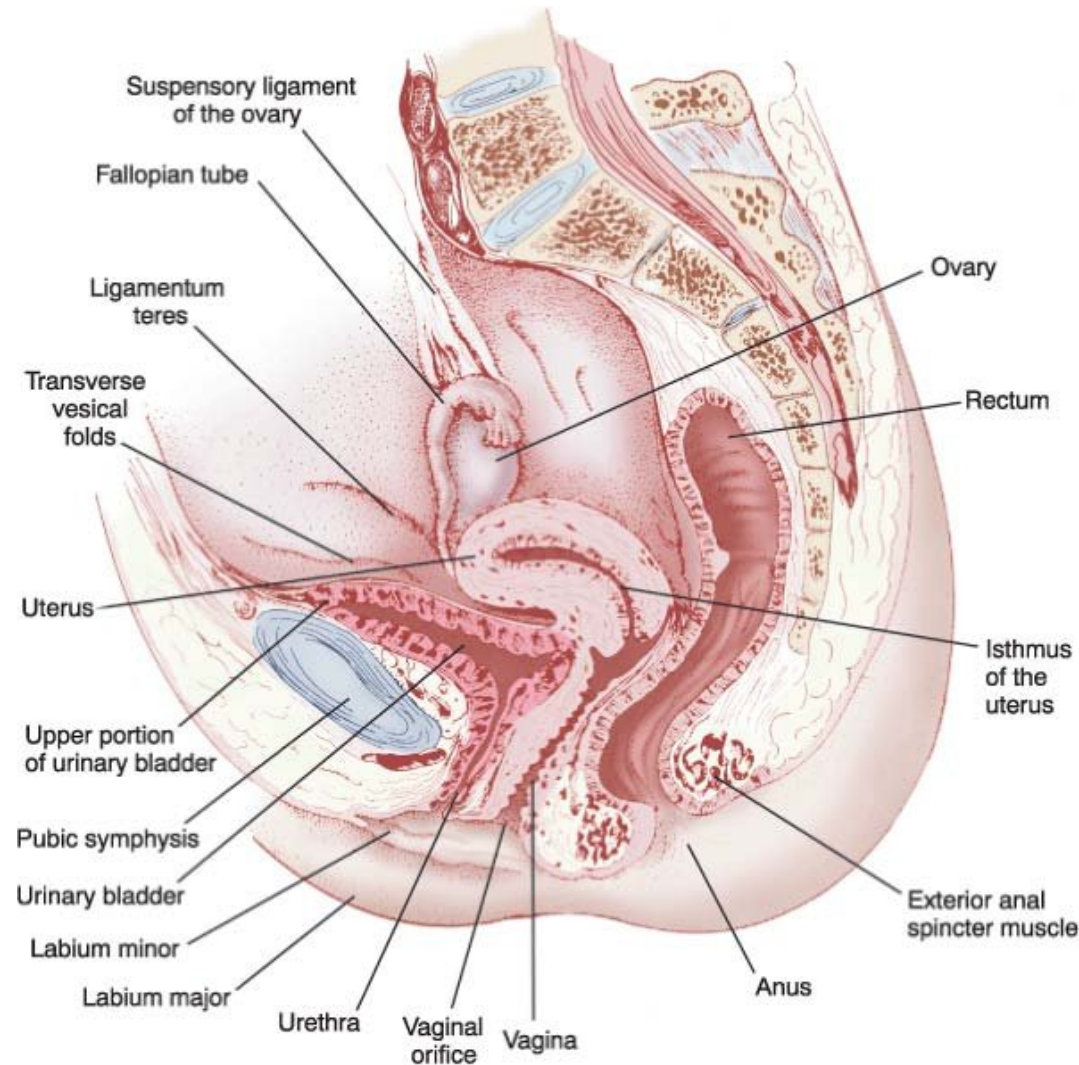
ANATOMY OF URINARY BLADDER



BY: M. G. HAMMAD

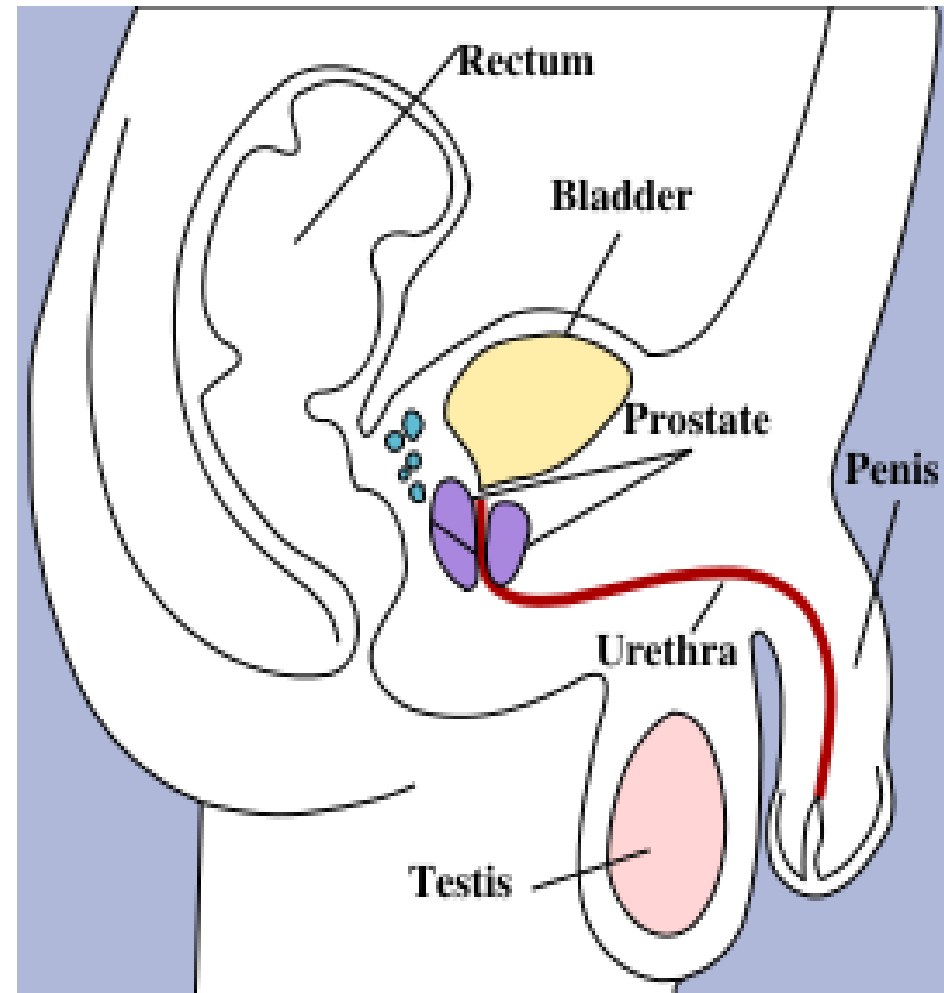
URINARY BLADDER

- It is a hollow muscular organ.
- Which serves as reservoir of urine.
- Discharge urine periodically.



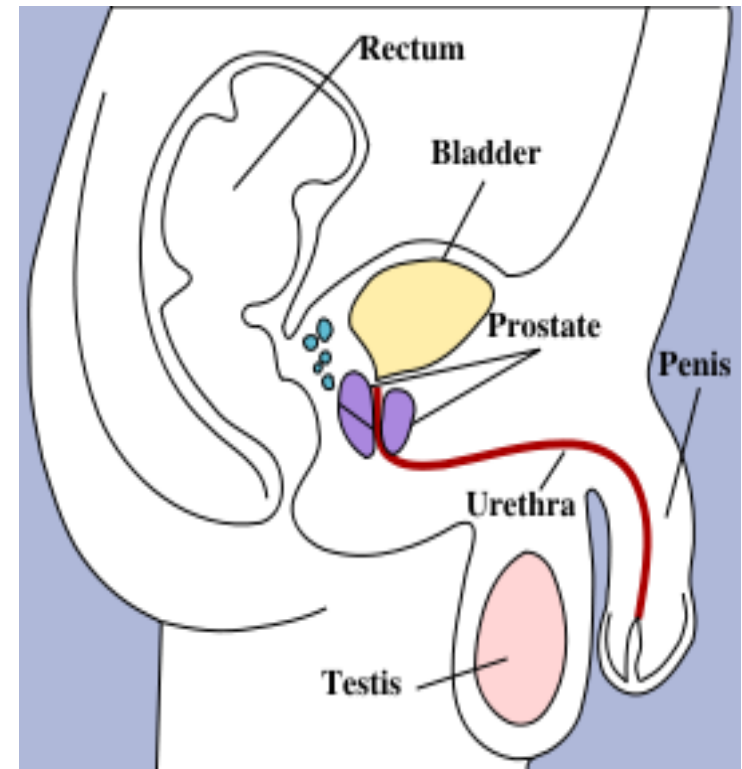
URINARY BLADDER:- LOCATION

- Empty bladder is in the anterior part of pelvic cavity.
- When it gets filled it becomes abdominopelvic organ.
- Reaching up to the umbilicus or even higher.
- Normal capacity of bladder is about **200-300 cc**.



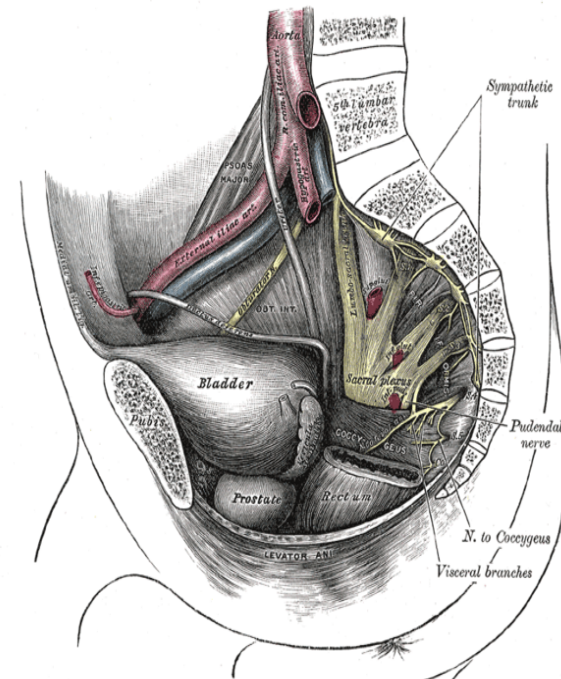
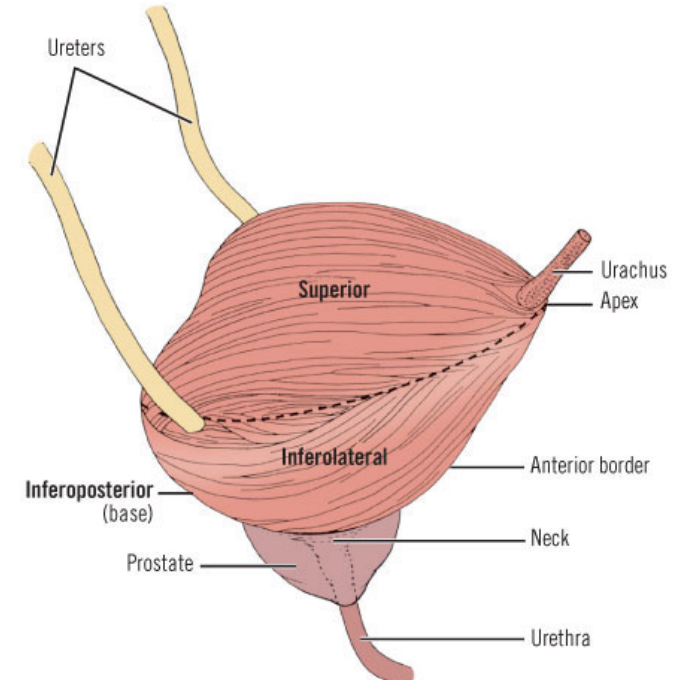
URINARY BLADDER:- CAPACITY

- Normal capacity of bladder in an adult is about **200-300 cc**.
- Filling of urine beyond 220 cc causes a **desire to micturate** and bladder is usually emptied when filled about 250 cc.
- Filling up to 500 cc is tolerated but beyond this becomes **painful**.
- Referred pain is felt in the lower part of **anterior abdominal wall, perineum and penis**.



URINARY BLADDER:- PARTS

- Empty bladder is pyramidal in shape.
- It has:
 - ✓ 4 Angles:
 - Apex.
 - Base.
 - Two lateral angles.



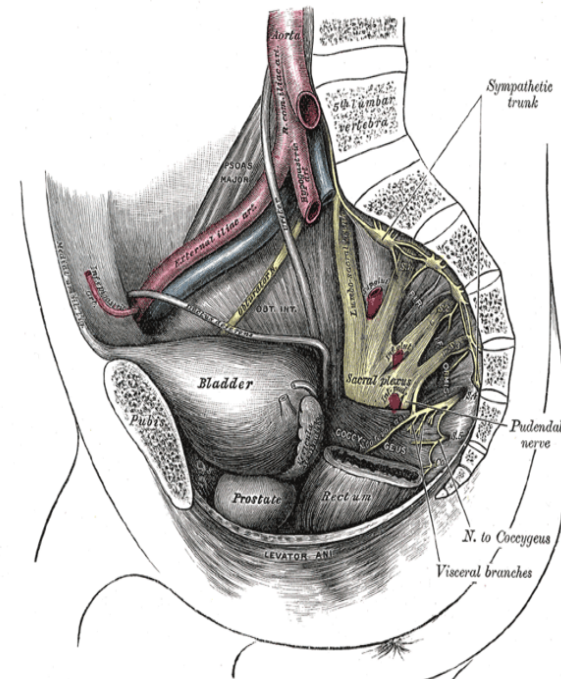
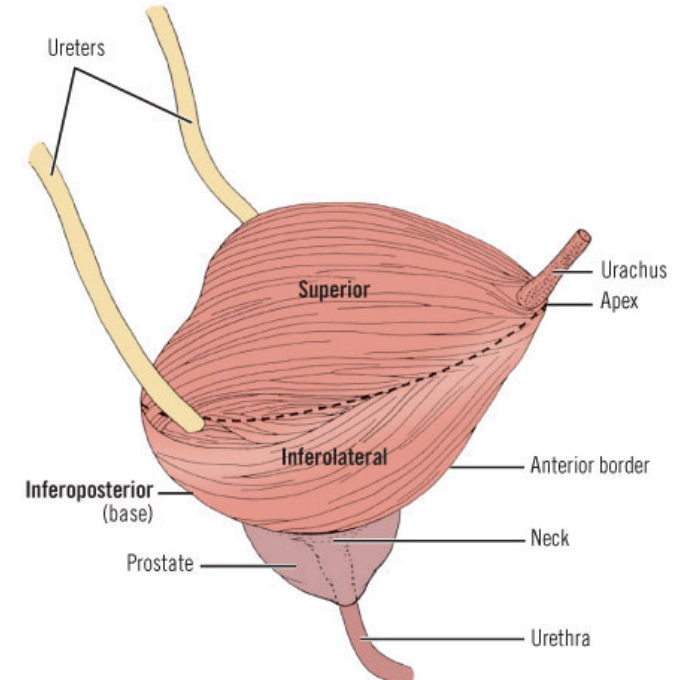
URINARY BLADDER:- PARTS

✓ 4 Surfaces:

- Superior.
- Posterior.
- Inferolateral {R- L}.

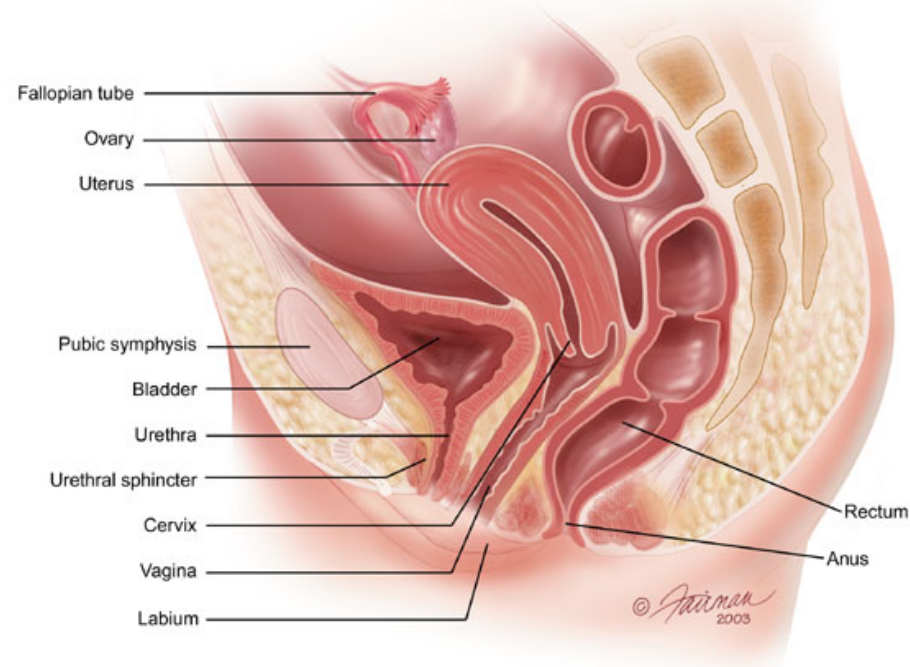
✓ 4 Borders:

- Posterior.
- Anterior.
- Lateral borders {R- L}.
- Full bladder is oval in shape.



URINARY BLADDER:- RELATIONS

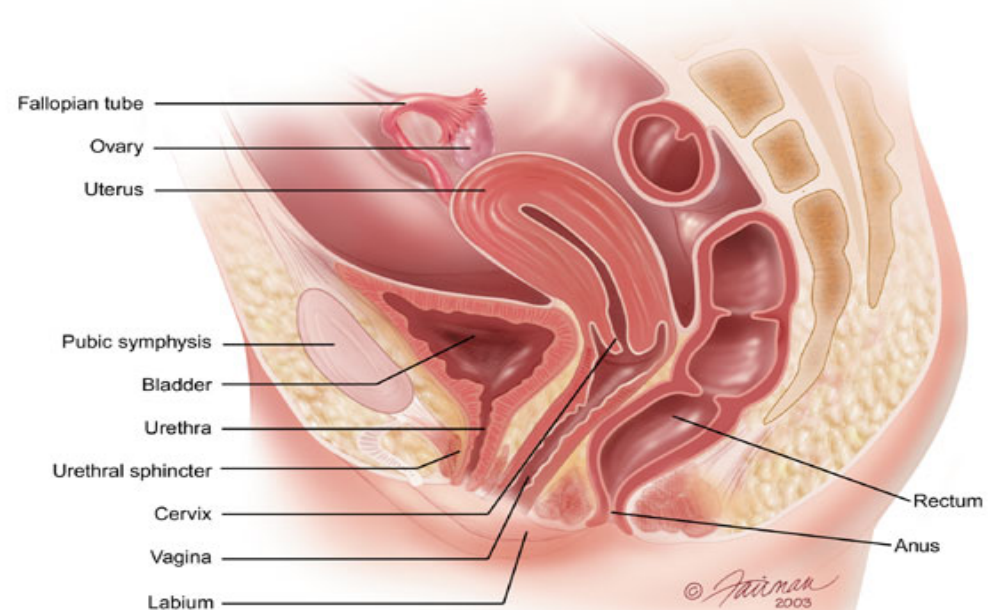
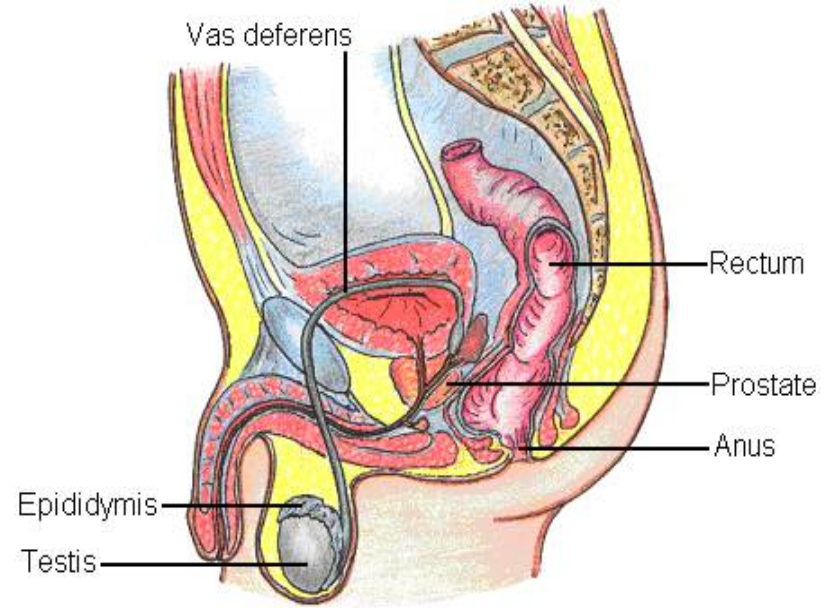
- ❑ Apex is connected to umbilicus by urachus (obliterated allantois diverticulum).
- ❑ Neck is continuous with urethra. In males it rests on upper surface of prostate.
- ❑ In females to urogenital diaphragm.
- ❑ Lateral angles are pierced by ureters.



URINARY BLADDER - RELATIONS

❑ Superior surface:

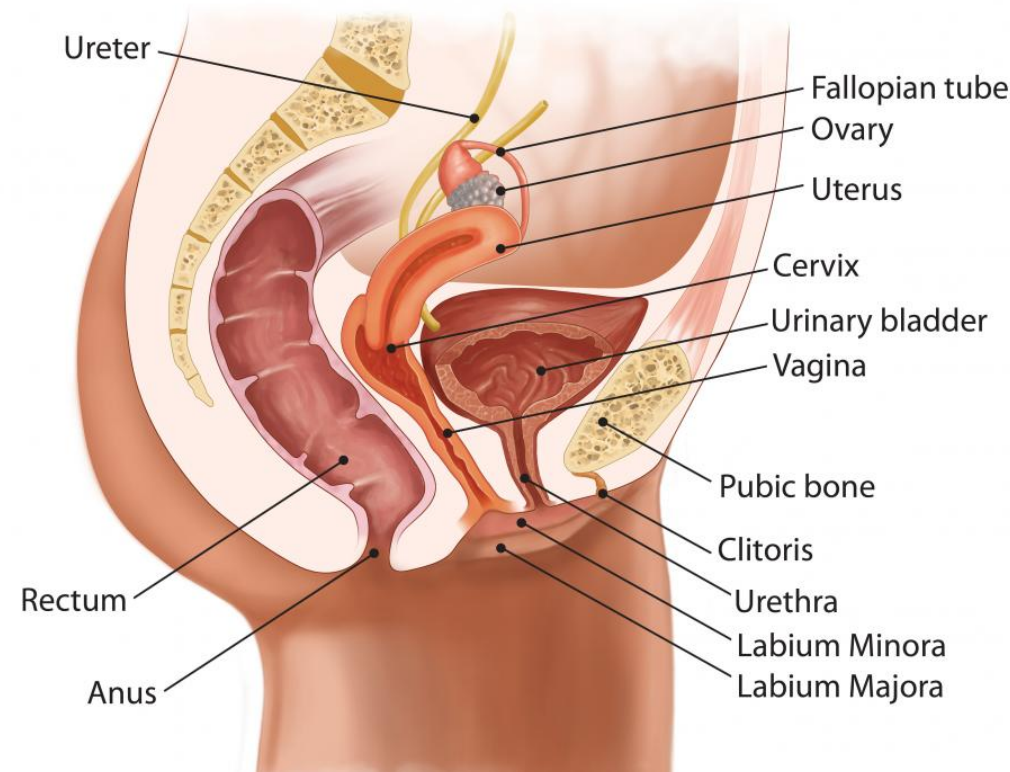
- ✓ In males: entire superior surface is covered by peritoneum.
- ✓ In females:
 - Anterior 2/3 is covered by peritoneum.
 - Posterior 1/3 is not covered by peritoneum and related to supravaginal part of cervix.



URINARY BLADDER – RELATIONS

❑ Inferolateral surfaces:

- ✓ Are related pelvic floor.
- ✓ Between bladder and pubis there is a space called retropubic space filled with retropubic fat.



URINARY BLADDER – RELATIONS

- Posterior surface:

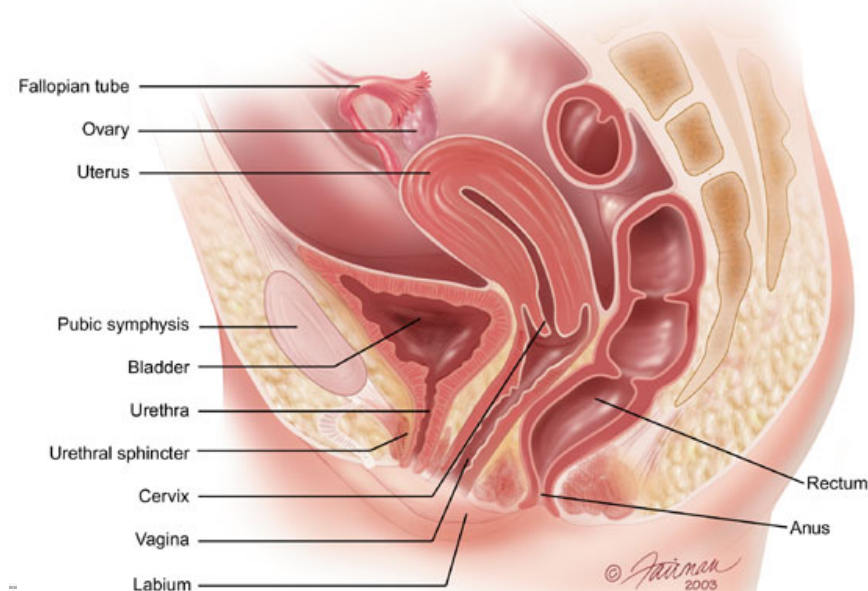
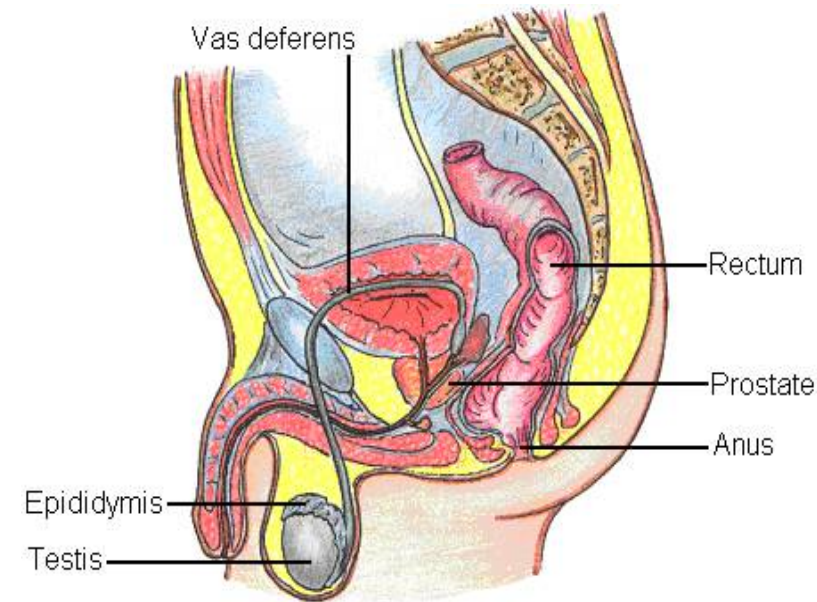
- In male:

- ✓ Upper part of this surface is covered by peritoneum and related to rectovesical pouch.

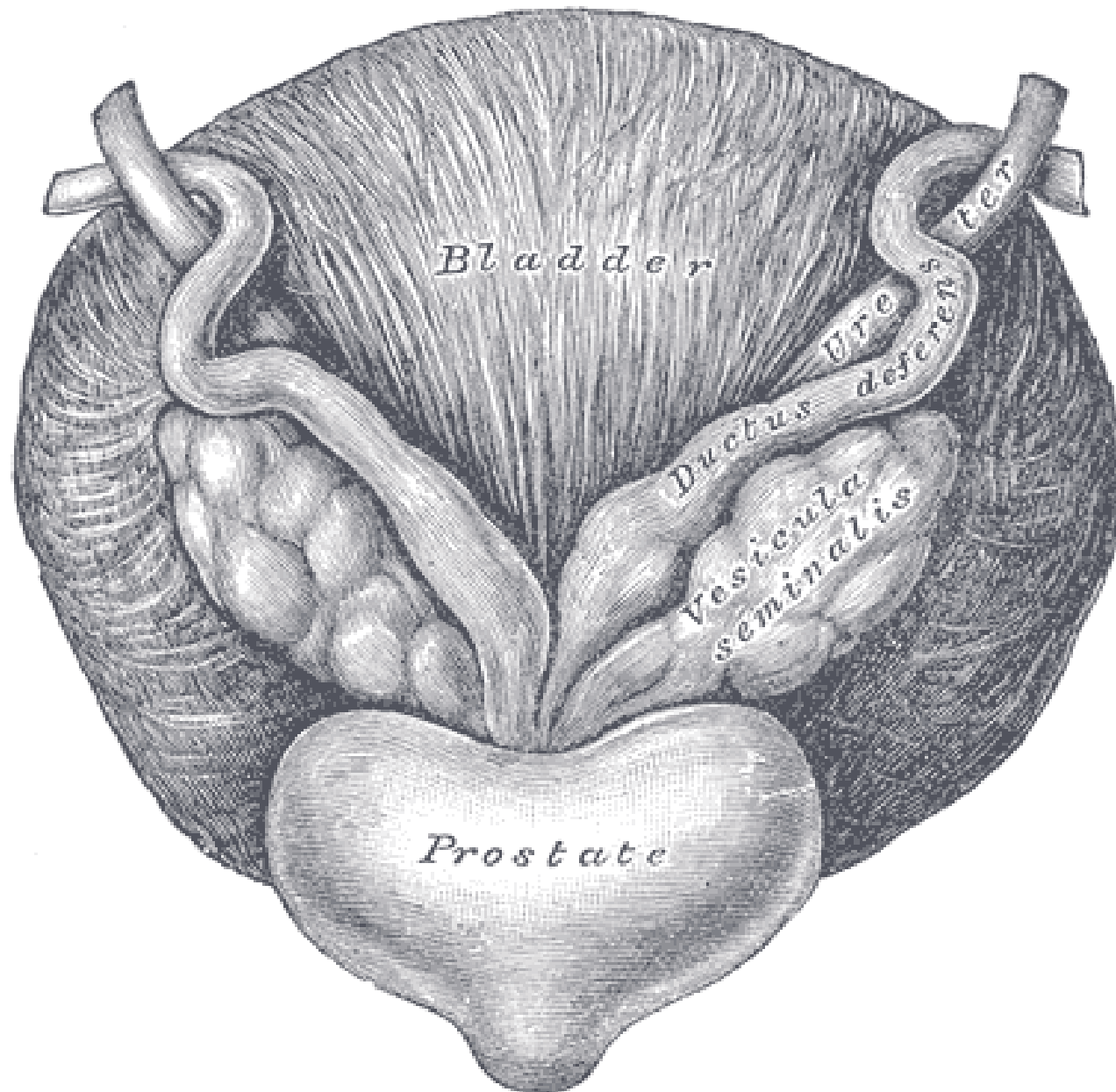
- ✓ Lower part is not covered by peritoneum and related to seminal vesicle and vas deferens.

- In female:

- ✓ Whole of this surface is related to anterior wall of vagina.

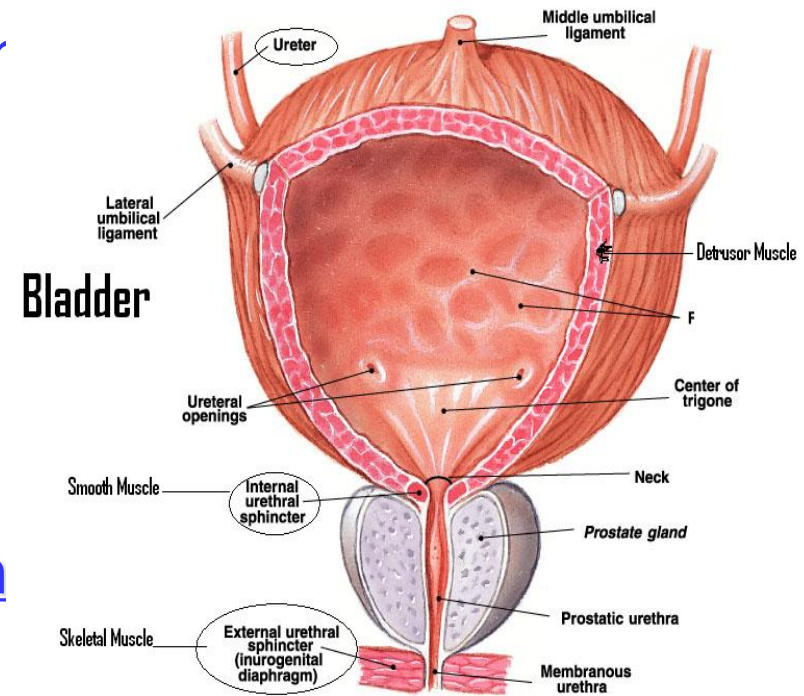


URINARY BLADDER – RELATIONS



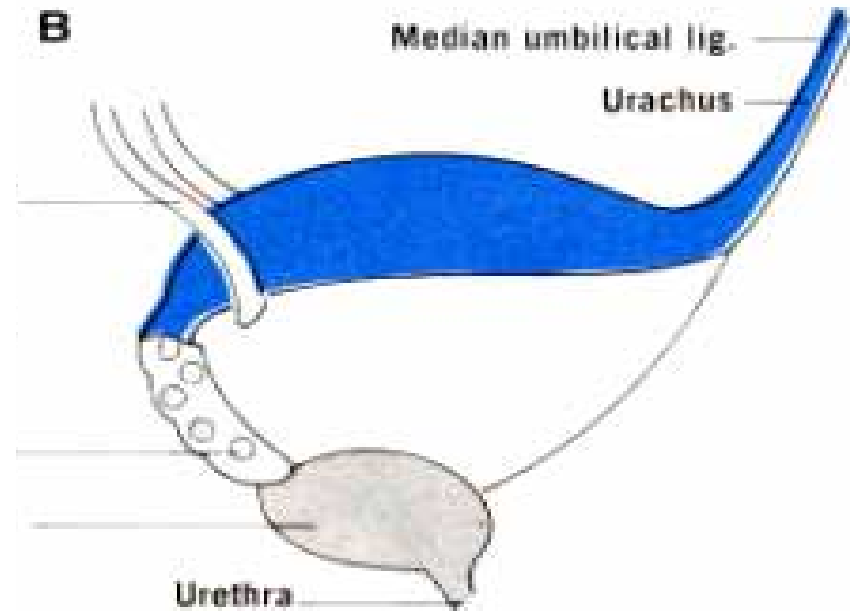
INTERIOR OF BLADDER

- The mucous is thrown into folds {disappear when the bladder is full}.
- Small triangular area over the lower part of base of bladder, mucosa is smooth (r folds) due to its firm attachment to muscular coat.
- This area is called TRIGONE.
- It is pink in color.
- In the upper lateral angles ureters open
- In the inferior angle there is internal urethral orifice.
- Uvula vesicae is a slight projection posterior to urethral orifice formed by middle lobe of prostate.



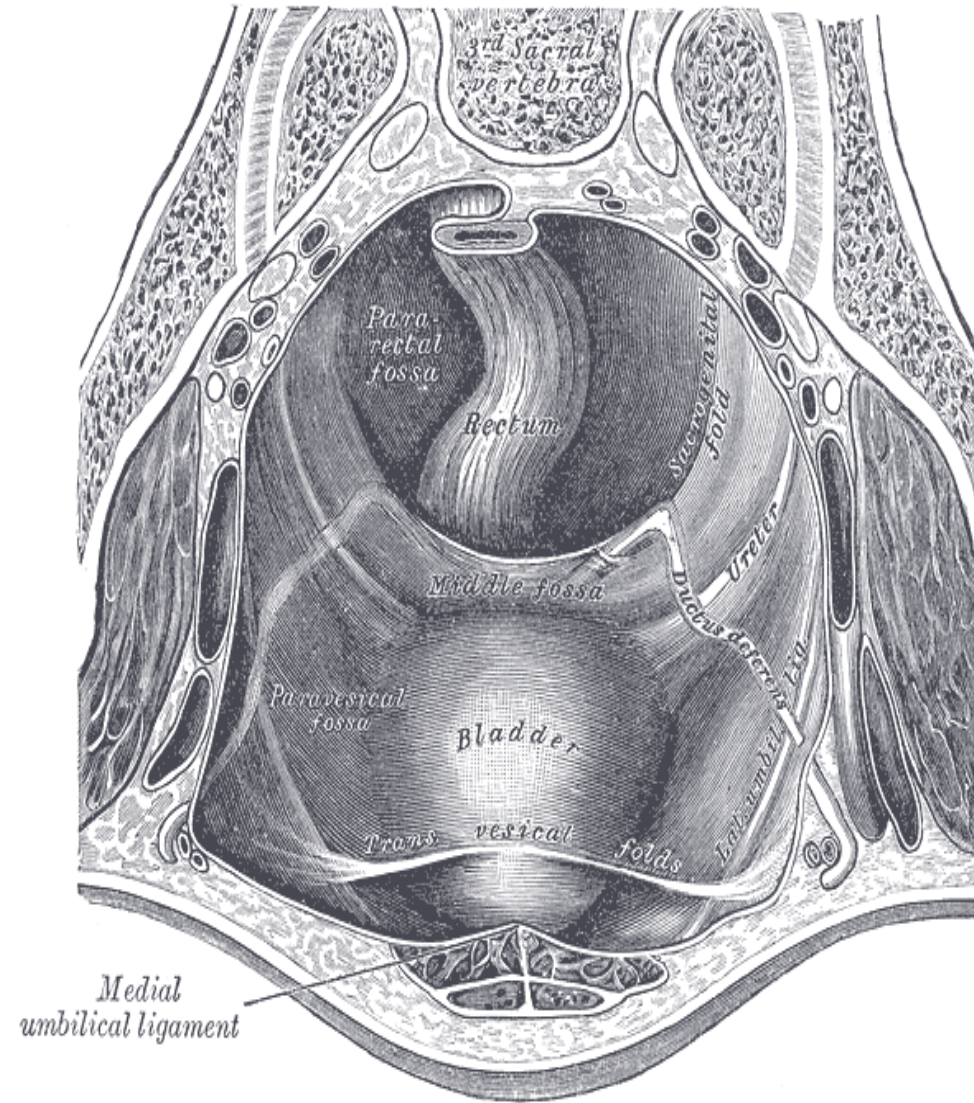
LIGAMENTS & SUPPORT OF BLADDER

- Ligaments support the bladder are two types:
 - ✓ True ligaments.
 - ✓ False ligaments.
- True ligaments:
 - Thickened pelvic fascia.
 - They are:
 - ❖ Lateral true ligament: attached from side of bladder to lateral pelvic wall.
 - ❖ Posterior true ligament: attached from lateral borders of base of the bladder to lateral wall
 - ❖ Medial & lateral puboprostatic ligaments: (pubovesical in female): attached from body of pubis to prostate.
 - ❖ Median umbilical ligament: formed by urachus.



LIGAMENTS & SUPPORT OF BLADDER

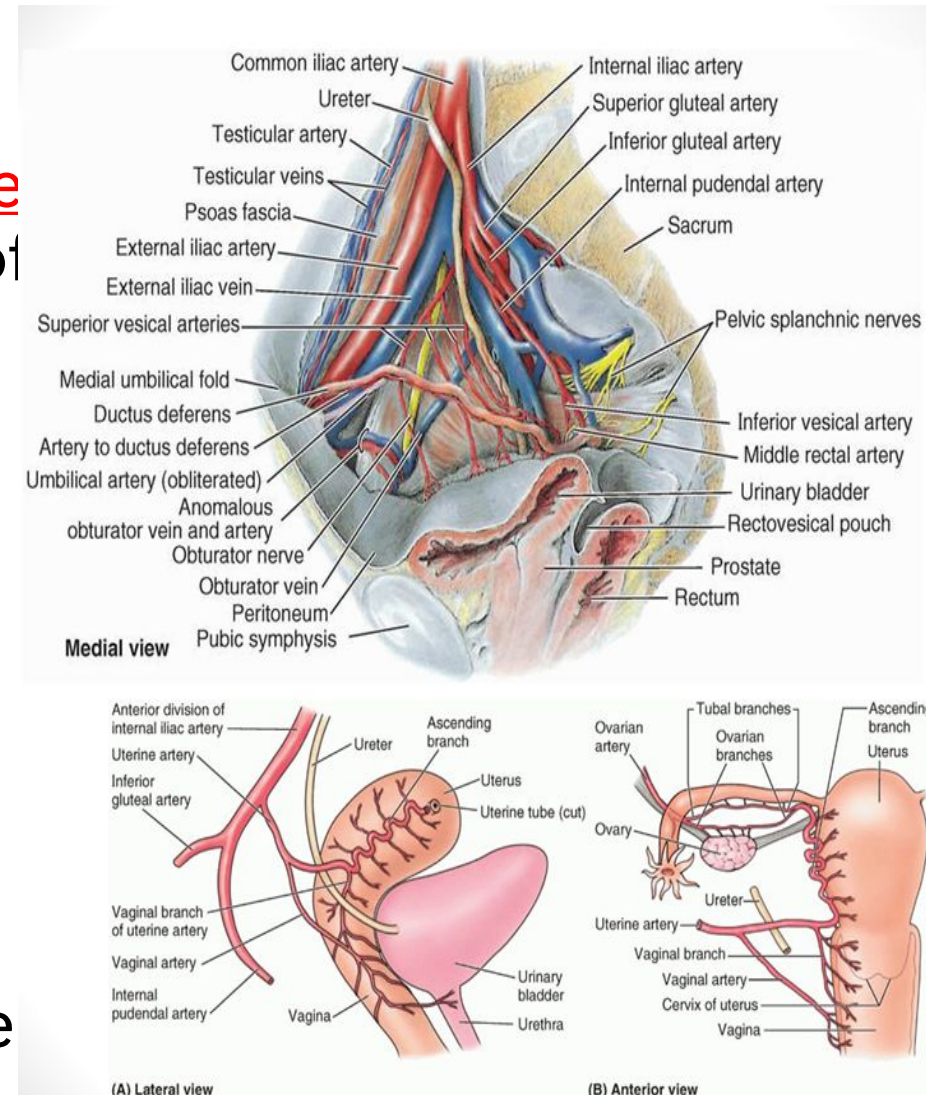
- False ligaments:
- ✓ Peritoneal folds.
- ❖ Median umbilical fold: peritoneal fold raised by urachus.
- ❖ Two medial umbilical folds: raised by obliterated umbilical arteries.
- ❖ Lateral false: peritoneal fold raised by lateral true ligaments.
- ❖ Posterior false: peritoneal fold raised by posterior true ligament.



URINARY BLADDER - BLOOD SUPPLY

 Arterial supply:

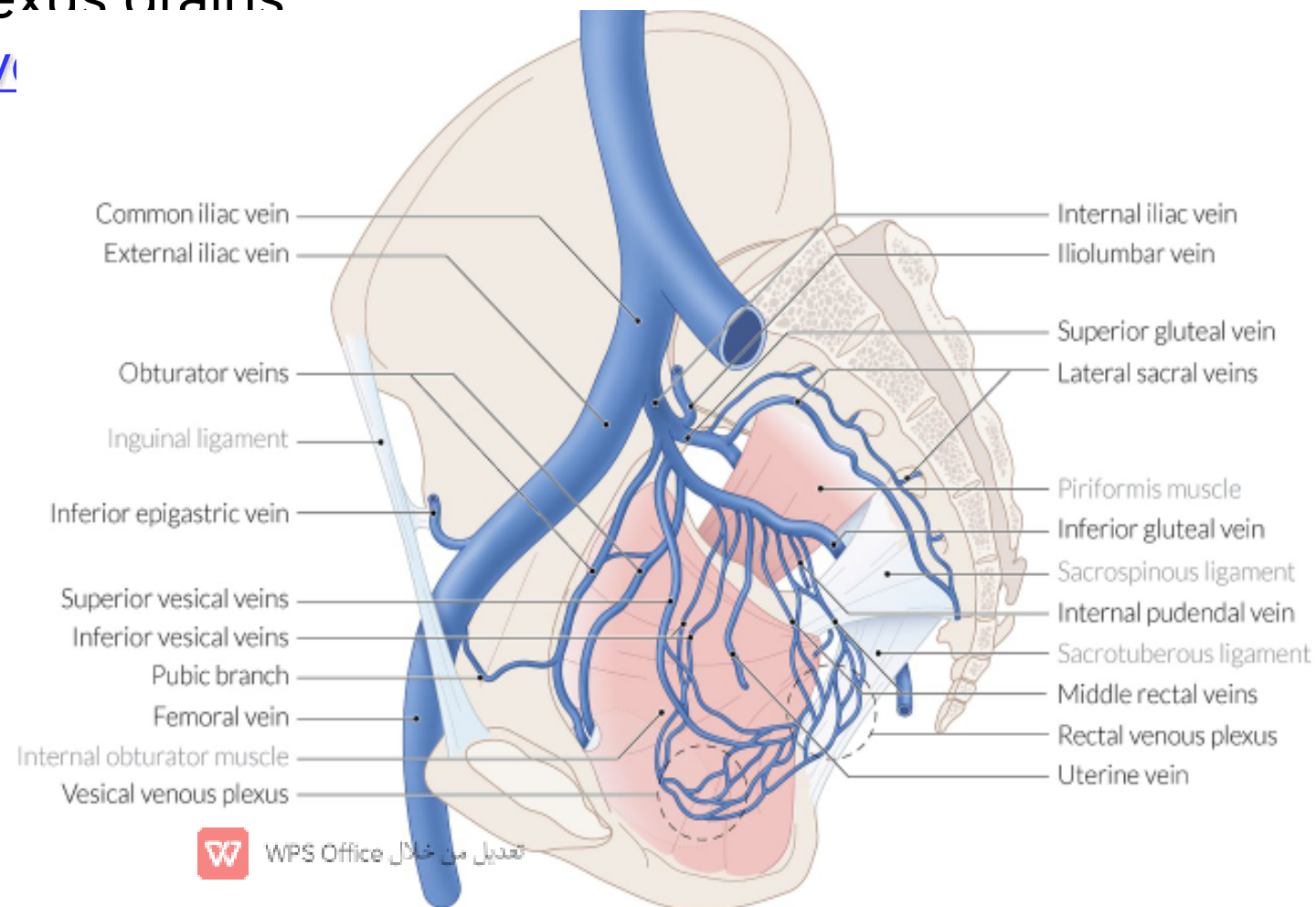
- Main arterial supply:-
- ✓ Superior and Inferior Vesical arteries branches from the anterior trunk of internal iliac arteries.
- Additional arterial supply:-
- ✓ Obturator artery.
- ✓ Inferior gluteal arteries.
- ✓ In the female Uterine and Vaginal arteries.
- The inferior vesical artery in female called VAGINAL ARTERY.



URINARY BLADDER – BLOOD SUPPLY

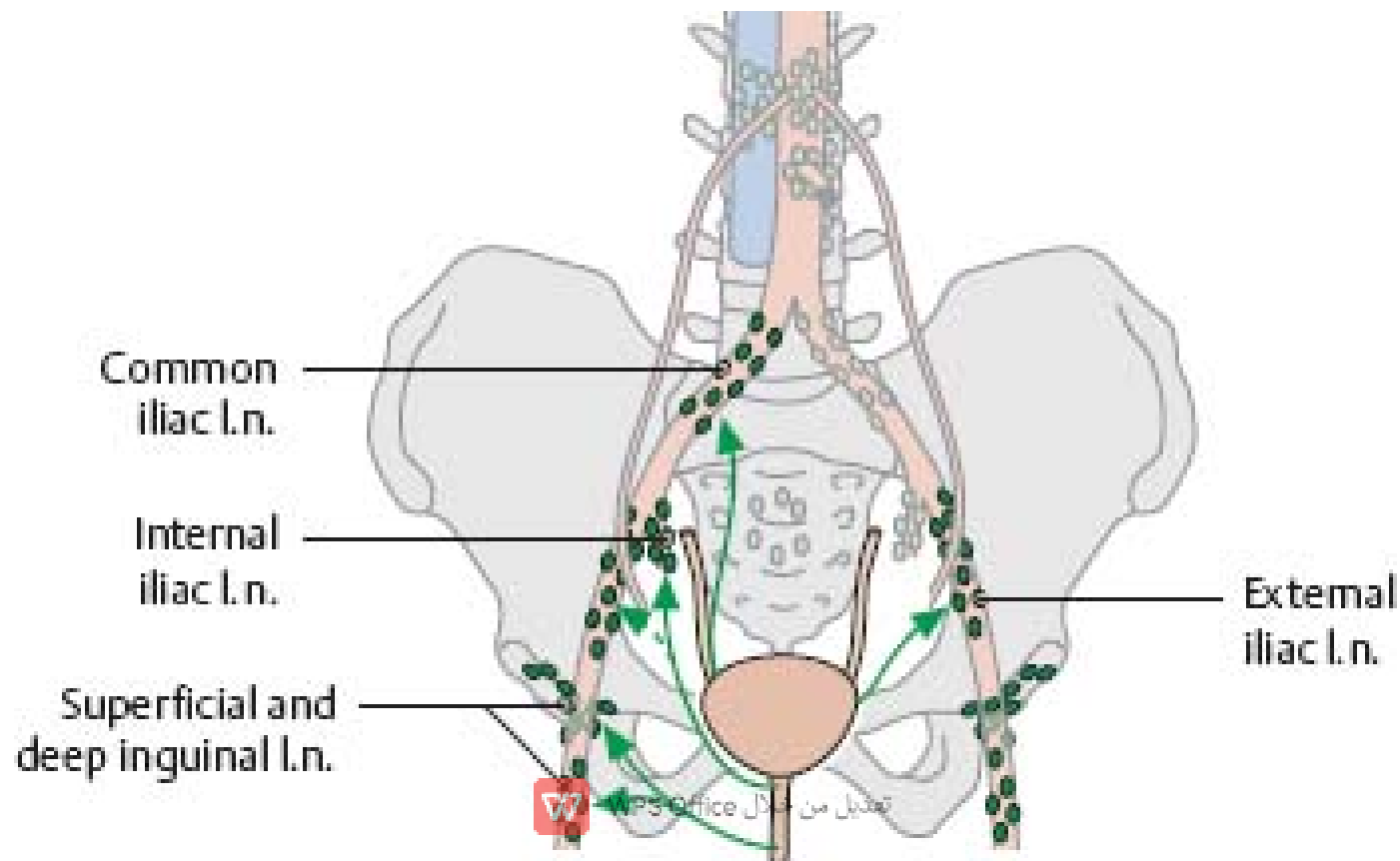
□ Venous drainage:

- Drained by vesical venous plexus lying on the inferolateral surface.
- Veins from this plexus drains into internal iliac v



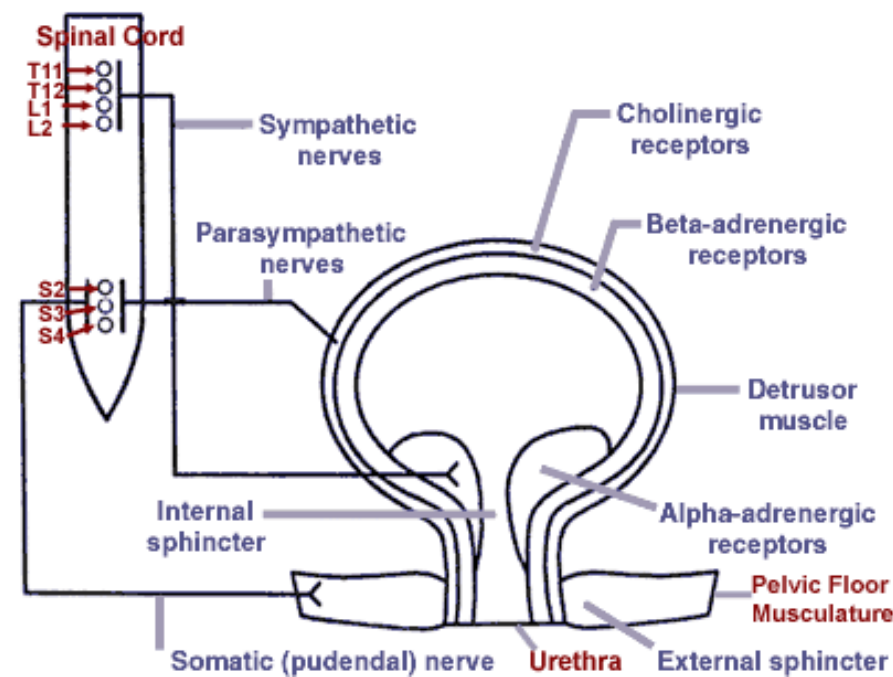
URINARY BLADDER – LYMPHATIC DRAINAGE

- Mainly drain into external iliac lymph nodes.
- Some to internal iliac lymph nodes.



URINARY BLADDER – NERVE SUPPLY

- Vesical nerve plexus consists of sympathetic and parasympathetic component.
- Parasympathetic from (nervi erigentes) S2, 3, 4: supplies motor fibers to detrusor muscle and inhibitory to the sphincter vesicae.
- Sympathetic nerves supply motor fibers to sphincter vesicae and inhibitory to the detrusor muscle.
- Voluntary motor control of external sphincter is by pudendal nerves.



THANK YOU

